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CipherDocs: A Secure AI-Powered Solution to Prevent Forgery of Critical and Official Documents Using Polygon Blockchain-Based Encryption and Verification

B.Tech Project

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Introduction

Nowadays, many documents in education, government, and businesses are stored digitally instead of on paper. This has increased the need for fast and reliable document verification. However, online document sharing has also increased the risk of fake or modified documents, making trust and authenticity very important.

Problem Statement

Document forgery and tampering are becoming major challenges in modern digital systems. Centralized storage solutions are vulnerable to security breaches, while manual document verification is time-consuming, inefficient, and prone to human error. In addition, the rise of AI-generated fake documents has made detecting manipulated content even more difficult. Therefore, there is a strong need for a secure, tamper-proof, and intelligent document verification system that ensures authenticity, integrity, and trust.

Motivation

- To eliminate document forgery using decentralized trust mechanisms
- To ensure long-term integrity of issued digital documents
- To reduce dependency on manual and human-driven verification
- To enable instant, reliable third-party verification
- To preserve user privacy during authentication
- To design a scalable and practical real-world solution

Objectives

- Use Polygon blockchain for decentralized, low-cost, and scalable document verification
- Ensure document integrity and tamper detection using SHA-256 hashing
- Secure documents using AES-256 encryption and IPFS-based decentralized storage
- Enable dynamic QR-based real-time document verification
- Implement smart contract-based issuance, verification, revocation, and expiry tracking
- Incorporate AI-assisted OCR, NLP, and VLM-based validation to detect textual, semantic, and structural irregularities

Alignment with Sustainable Development Goals (SDGs)

- SDG 9 — Industry, Innovation and Infrastructure
    - It focuses on innovation and secure digital infrastructure. Our blockchain-based system provides secure digital document verification.
- SDG 16 — Peace, Justice and Strong Institutions
    - It focuses on justice and transparency. Our system helps prevent fake documents and increases trust.
- SDG 10 — Reduced Inequalities
    - It supports equal access because anyone can verify documents remotely from anywhere.

Literature Survey

CipherDocs

Sr. No.	Paper Name	Core Technology	Specific Limitations Identified	CipherDocs Solution / Novelty
1	Certifier dapp-decentralized and secured certification system using blockchain	Ethereum DApp	High Gas Fees: Ethereum dependency makes it costly. No AI: Purely hash-based verification.	Polygon Network: Reduces fees significantly. AI Integration: Adds semantic forgery detection.
2	DOC-BLOCK: A blockchain based authentication system for digital documents	Ethereum + IPFS	Binary Validation: No explanation for failure. Privacy Leaks: Public IPFS content visibility.	Explainable AI: Provides reasons for rejection. Context-Aware Encryption: Secures off-chain data.
3	A Comparative Study of Ethereum and Polygon for Implementing NFT-Based Certification Systems	Eth vs. Polygon	L2 Security Trade-offs: Notes security variance. Complexity: Cross-chain management.	Native Polygon Deployment: Optimizes specifically for L2 speed/cost, accepting trade-offs for usability.
4	Docschain: Blockchain-based IoT solution for verification of degree documents.	IoT + OCR	OCR Fragility: Fails with poor scans/lighting. Format Limits: B&W only.	Multimodal AI: Robust to image noise/artifacts. Digital-First: Focuses on born-digital assets alongside scans.

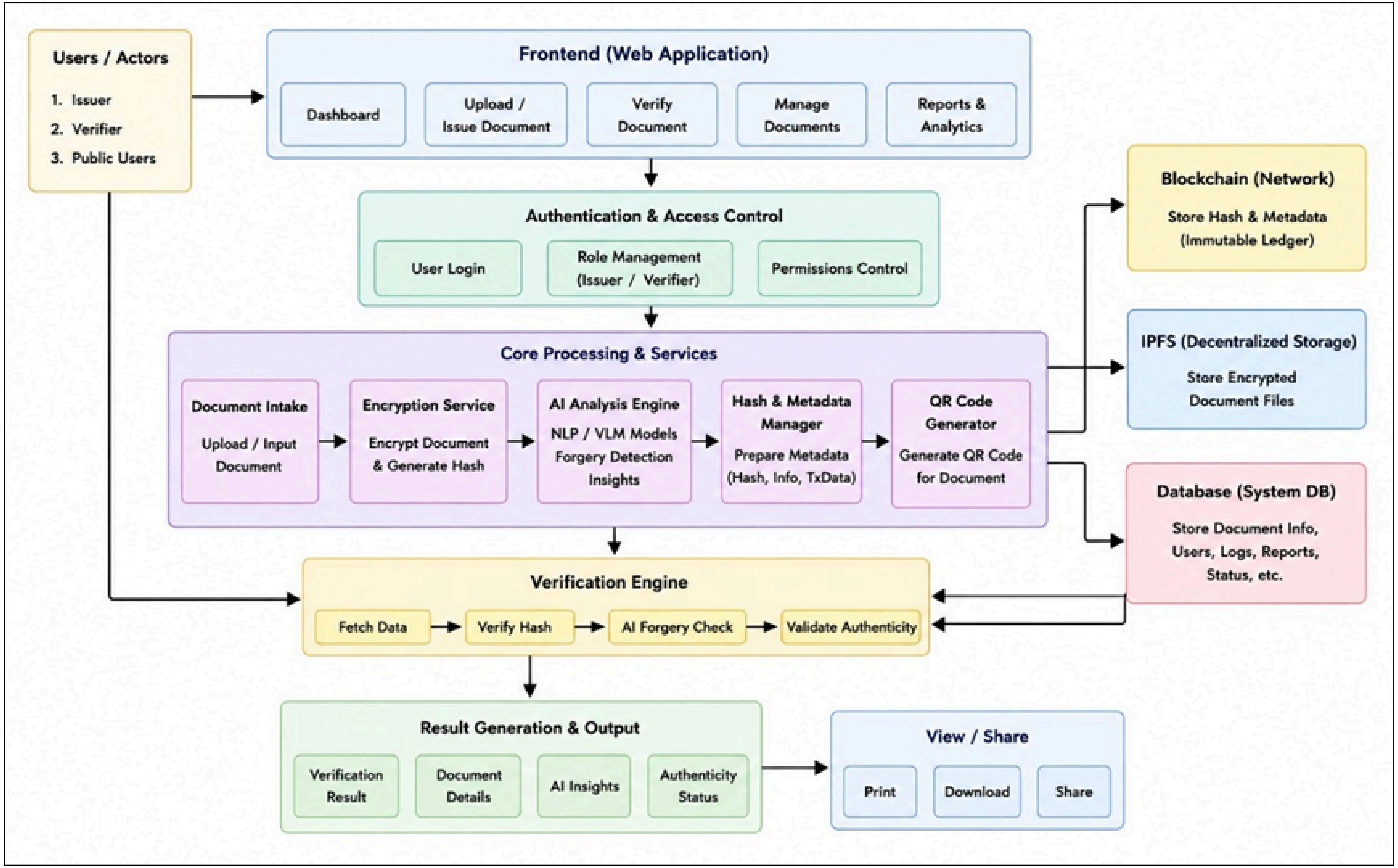
Gap Analysis

CipherDocs

- Existing systems using Ethereum have very high transaction costs, so large organizations may find them expensive to use.
- Traditional systems use static QR codes and manual checking, so real-time verification is not possible.
- Most systems only compare document hashes and cannot detect intelligent or meaningful changes in documents.
- There is no proper automation for document expiry, renewal, or revocation.
- Existing systems also do not properly separate roles between issuers, document owners, and verifiers.

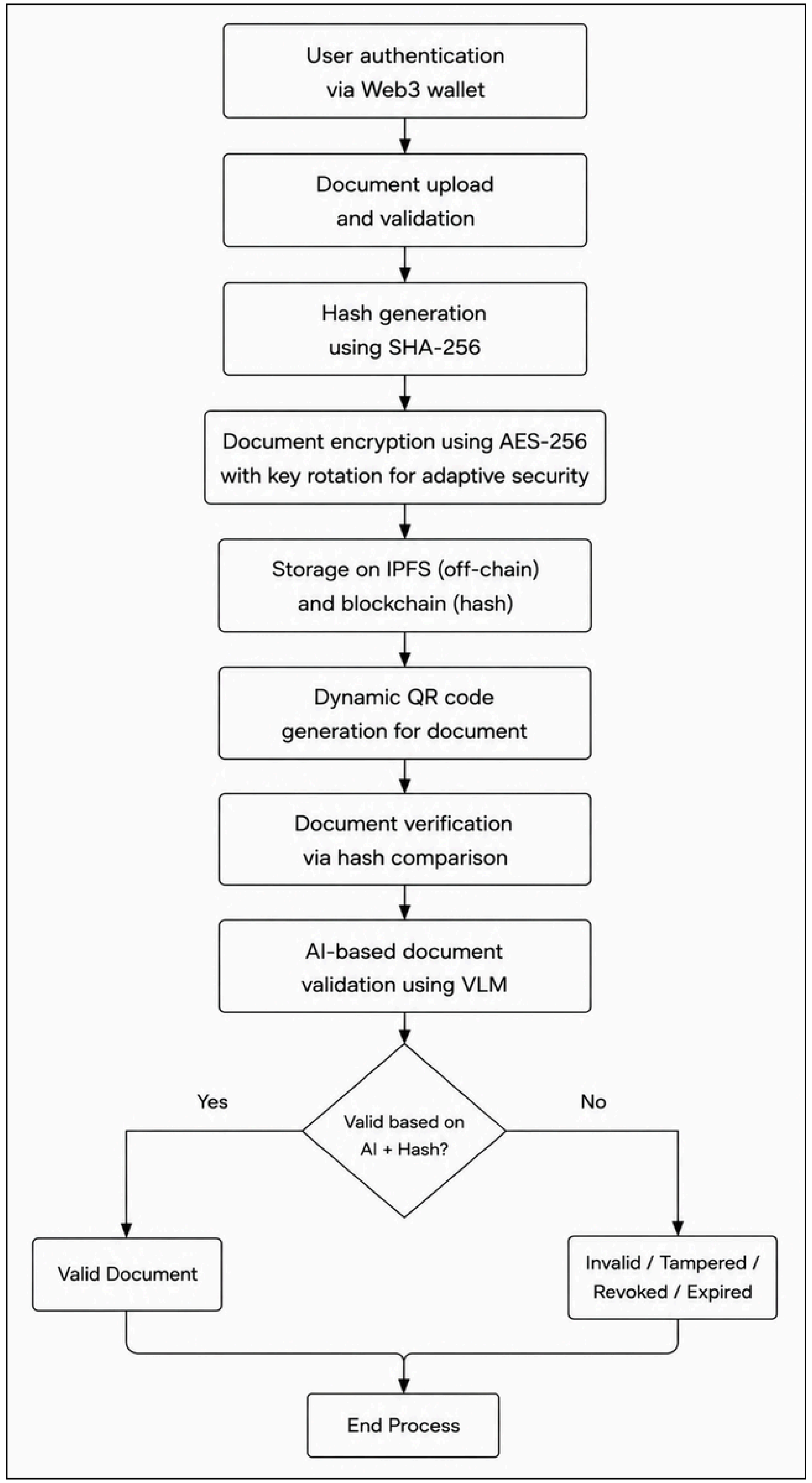
System Architecture

CipherDocs



Workflow

- User authentication via Web3 wallet
- Document upload and validation
- SHA-256 hash generation
- AES-256 encryption with adaptive key rotation
- Encrypted document storage on IPFS (off-chain)
- Document hash storage on Polygon blockchain
- Dynamic QR code generation
- AI-based document validation using VLM
- Document verification via hash comparison
- Display result (Valid / Invalid / Revoked / Expired)



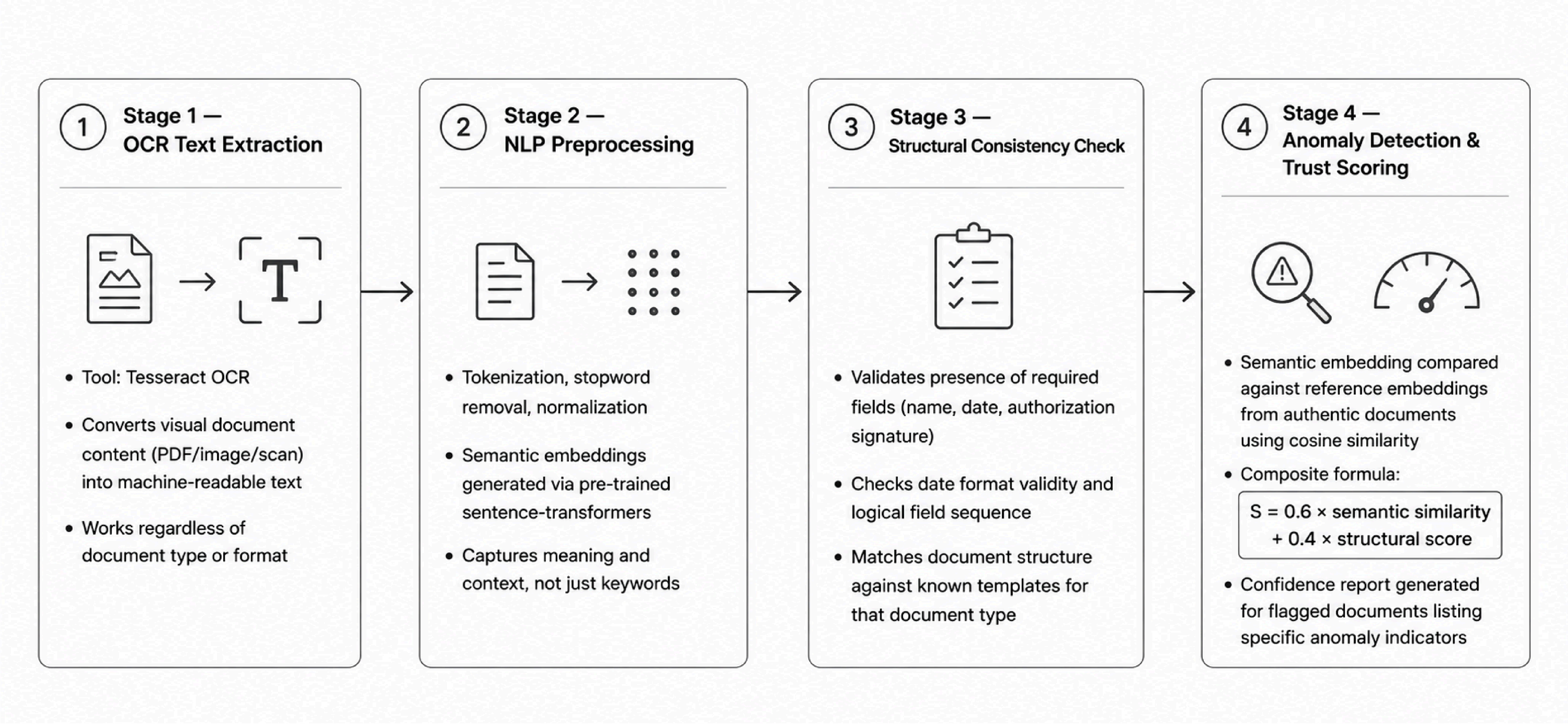
AI Validation Module: Overview & Purpose

CipherDocs

- Hash verification detects exact tampering only
- Cannot detect semantic/content-level manipulation
- AI adds an intelligent second verification layer

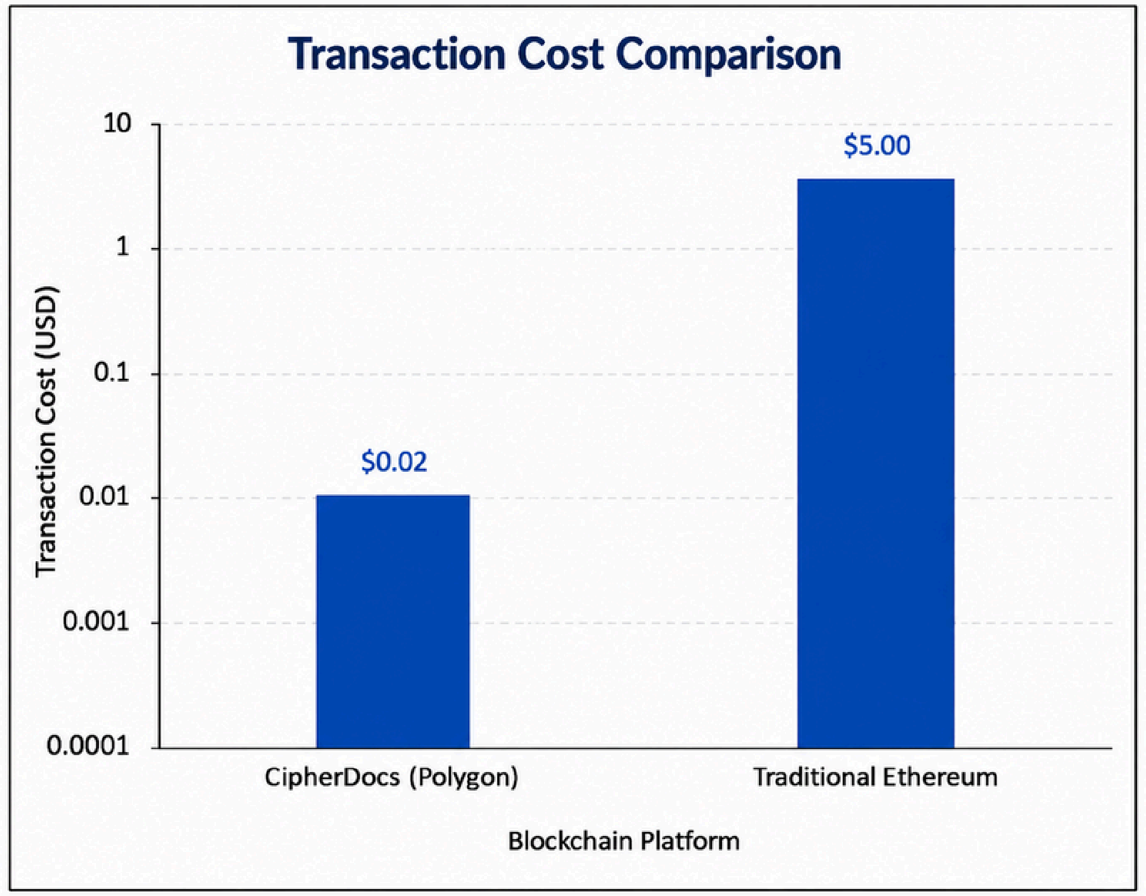
- AI Validation Module
    - OCR extracts text from documents
    - NLP checks semantic consistency
    - Structural validation of fields & formats
    - Generates Trust Score ($S \in [0,1]$)

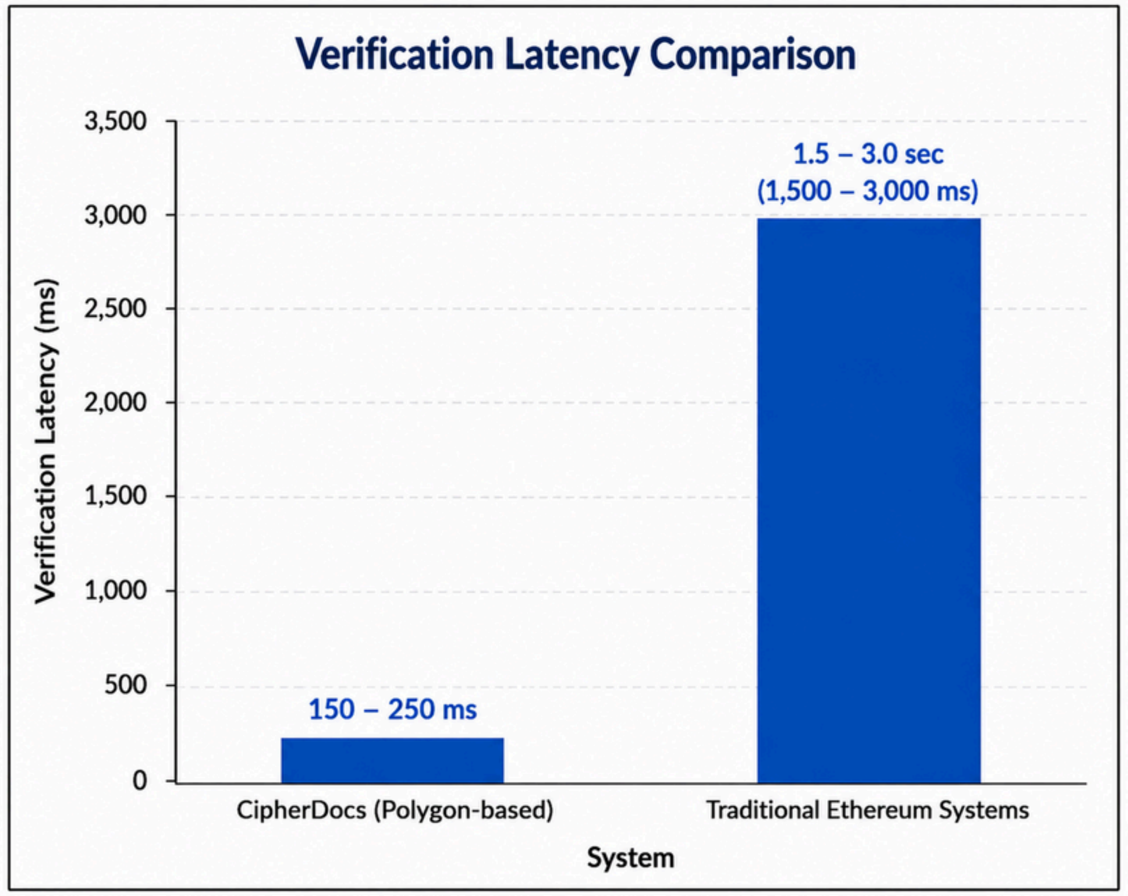
AI Pipeline: 4-Stage Architecture



Performance Metrics

- Ultra-Low Cost: ~$0.002 average transaction cost
- High Scalability: Supports large-scale users and documents
- Fast Verification: ~150–250 ms verification response
- High Throughput: Efficient parallel transaction processing
- Advanced Security: SHA-256, AES-256, blockchain immutability, and AI-based tamper detection
- Efficient Storage: IPFS-based encrypted off-chain storage





Results

Parameter	CipherDocs (Polygon)	Traditional Systems (Ethereum)	Bitcoin
Transaction Cost	\$0.0005 – \$0.02	\$0.2 – \$0.66	\$0.1987 – \$0.77
Throughput (TPS)	~180 TPS	~15 – 30 TPS	~6 – 10 TPS
Verification Latency	150 – 250 ms	1.5 – 3 sec	~20 – 40 min
Block Time	~2 sec	~12 sec	~10m
Scalability	High	Moderate	Low
Storage Method	IPFS + Blockchain Hash	Mostly On-chain / Centralized	UTXO model
Security	AES-256 + SHA-256 + Blockchain	Limited to Blockchain Only	Proof of work with longest-chain security
QR Verification	Dynamic QR Verification	Limited / Static QR Verification	Supports BIP-21/BIP-321 URI QR payment links

Certificate is valid and authentic. VERIFIED

ISSUER WALLET ADDRESS
0xcae0a1144fe58397795e422786f076fa9c5a721c

USER WALLET ADDRESS
0xfD771b82c234222Fb00b5a86a5800348f114e8BD

CERTIFICATE ISSUED AT
4/28/2026, 12:09:50 AM

BLOCKCHAIN TX
0xb6b4c67c2cf691f093b02e630b0b1126a4dc082f6593d3dc13d2a8ad904b7b53

Document has been tampered. TAMPERED

Uploaded document does not match the originally issued certificate.

VERIFIED FILE
sy-fake.pdf 210.4 KB

Certificate has been expired. EXPIRED

Certificate validity has expired.
Expired On: 1/21/1970

VERIFIED FILE
Rent Agreement.pdf 148.5 KB

Certificate has been revoked. REVOKED

Certificate has been revoked.
Revoked At: 1/21/1970, 7:19:03 PM
Expired On: 1/21/1970

VERIFIED FILE
Land Record.pdf 2.3 KB

AI Trust Score

Using the uploaded verification document to calculate trust score. Our AI will analyze semantic similarity, structure, and metadata to calculate a trust score (0-100).

100
HIGH Trust Level
✓ Document appears authentic

100	100	100
Text Similarity	Structure	Metadata
Semantic content match	Format & layout match	Properties match

Detailed Analysis
The document shows high authenticity.

MetaMask

Issuer

Transaction request

Network: Amoy
Request from: cipherdocs.vercel.app
Interacting with: 0x9d368...Aa467

Network fee: < $0.01 POL
Speed: Site suggested ~2 sec

AI-Powered Analysis TAMPERED

Summary

The uploaded certificate has been marked as tampered by the cipherdocs verification system. This indicates that the certificate has undergone modifications after it was issued.

Verification Status

The status returned by cipherdocs is "tampered", suggesting that the uploaded certificate does not match the original certificate stored in the system.

Document Analysis

A comparison of the uploaded certificate text with the original certificate text stored in cipherdocs reveals several discrepancies. While the overall structure and content of the two certificates are similar, there are notable differences in certain fields.

Detected Differences

The following differences were detected between the uploaded and original certificates:

- The "ABC ID" field has been modified from "948128680857" to "949128680857".
- The "Year" field in the course "BIT4401" has been changed from "3" to "2".
- The "Course Name" for course code "BIT3401" has been altered from "DISCRETE MATHEMATICS" to no matching course, but "BIT3401" is associated with "LOGIC DESIGN AND COMPUTER ORGANIZATION" in the original, and "DISCRETE MATHEMATICS" is found in the uploaded text but associated with a different course code in the original.
- The "Cumulative CGPA" has been changed from "7.6" to "8.0".

AI-Powered Validation VERIFIED

Summary

The uploaded certificate has been verified against the original certificate stored in cipherdocs, and the result indicates that the certificate is valid.

Verification Status

The status returned by cipherdocs is "valid", indicating that the uploaded certificate matches the original certificate stored in the system.

Document Analysis

A thorough analysis of the uploaded certificate text and the original certificate text stored in cipherdocs has been conducted. The comparison reveals that the two documents are identical, with no discrepancies or modifications detected.

Conclusion

Based on the analysis, it can be concluded that the uploaded certificate is authentic and has not been tampered with. The certificate matches the original certificate stored in cipherdocs, and all details, including student information, course grades, and semester records, are consistent. Therefore, the document trust status is confirmed to be valid.

Research Paper I

CipherDocs

- Focuses on a decentralized and secure blockchain-based document verification framework
- Uses Polygon blockchain for low-cost and high-speed transactions
- Integrates IPFS for secure decentralized document storage
- Implements smart contracts for transparent issuance, verification, revocation, and expiry tracking
- Uses SHA-256 hashing and AES-256 encryption for document security and tamper detection
- Enables dynamic QR-based real-time document verification and authentication

Research Paper II

- Focuses on AI as an intelligent supportive layer for document validation
- Integrates an AI assistant for user guidance and verification support
- Uses OCR, NLP, and VLM-based analysis for contextual document validation
- Detects anomalies, inconsistencies, and forged document patterns
- Enhances verification accuracy, automation, and overall system usability

Conclusion

CipherDocs

- Proposed a decentralized framework to prevent digital document forgery
- Leveraged blockchain technology to ensure tamper-proof verification
- Integrated AI-based validation to enhance forgery detection
- Enabled quick and user-friendly verification using QR codes
- Demonstrated a scalable and cost-efficient solution using the Polygon blockchain

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Thank you!